

the evidence lower bound (ELBO), balancing reconstruction accuracy with a regularisation term that shapes a smooth latent space.

VAEs are generally stable and easier to train than GANs, and they provide explicit likelihood estimates that support tasks such as interpolation and anomaly detection. Their main limitation is output sharpness: generated images often appear blurrier compared to GAN results because of the trade-off between fidelity and regularisation. Even so, VAEs remain valuable where structured latent representations or probabilistic reasoning are important, particularly in scientific and hybrid generative systems.

**Diffusion models:** Diffusion models generate data by gradually denoising random noise, drawing inspiration from physical diffusion processes. They use two stages: a fixed forward process that incrementally adds noise to real data, and a learned reverse process that removes this noise step by step until structure reappears.

Formally, starting from a data sample  $x_0$ , the forward diffusion produces:

$$x_1, x_2, \dots, x_T,$$

where  $x_T$  approaches a Gaussian noise distribution. A neural network is trained to reverse this corruption by predicting the clean sample from a noisy input at time step  $t$ . To synthesize new data, the model begins with:

$$x_T \sim N(0, I) \text{ and applies } x_T \rightarrow x_{T-1} \rightarrow \dots \rightarrow x_0.$$

Because each denoising step is small and stable, diffusion models avoid the instability of adversarial training while producing highly detailed outputs. They currently achieve state-of-the-art results in image generation, often exceeding GANs in visual fidelity.

In practice, these models learn to transform pure noise into realistic samples, leading to strong diversity and quality, especially for images and

audio. Training is typically stable but computationally intensive. Their main drawback is slow sampling, since many iterative denoising steps are required, and they do not provide a clear low-dimensional latent space. Even so, diffusion models have become a central technique in modern generative AI due to their reliability and output quality.

Figure 2 illustrates a diffusion model that generates an image through iterative denoising. Step 1 starts with a random noise image. Steps 2 and 3 are intermediate steps, where the model removes noise and begins to reveal structure (in this toy example, a simple shape).

**Transformer-based generative models:** The Transformer is a deep learning architecture that reshaped generative modelling, particularly in natural language processing. Originally built for sequence tasks, its defining feature is the self-attention mechanism, which allows the model to consider relationships across an entire sequence at once, improving parallelization and long-range coherence.

Transformers power large language models such as GPT-3 and later systems. These models generate text autoregressively by predicting one token at a time while referencing the full prior context, enabling fluent and consistent long-form output. Beyond text, Transformers are also used in image, music, and code generation, often combined with other techniques such as diffusion conditioning or

discrete token modelling.

Transformer-based models excel at capturing global context and adapting to multiple tasks through prompting or fine tuning, exemplifying the ‘foundation model’ paradigm. However, they are resource intensive to train and run, and despite their fluency they can still produce inaccurate information. Their scale and complexity also make them difficult to interpret, which raises challenges for transparency and trust.

## Applications of generative AI

Generative AI is being applied across many domains, from language and visual media to audio and video. These systems demonstrate strong potential in automating creative work, personalising digital content, and supporting business and research activities.

### Text generation and language

**applications:** Natural language generation is one of the most mature areas of generative AI. Large language models such as GPT-3, GPT-4, and Google’s LaMDA can produce coherent and context aware text from short prompts. This capability supports several practical uses:

- **Content creation:** Assisting with articles, marketing copy, reports, and brainstorming drafts.
- **Conversational agents:** Powering chatbots and virtual assistants that answer questions and hold interactive dialogues.
- **Translation and summarisation:** Generating translations or concise summaries of long documents within

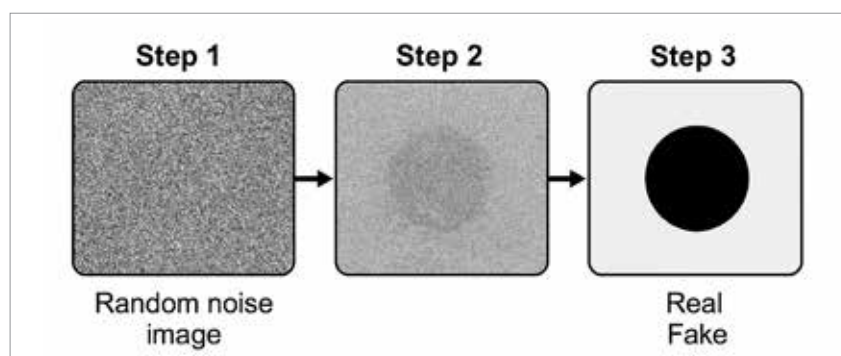


Figure 2: Illustration of a diffusion model